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Relationships Among Manual Body Functions, Manual Capacity, and Bimanual Performance Using the Prosthetic Upper Extremity Functional Index in Children With Congenital Hand Differences

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Background. Most surgical techniques intervene at the level of body functions of the upper limb, aiming to improve manual capacity and activity performance. However, the nature of the relationships among these levels of functioning and evidence for hand function variables predicting performance have scarcely been investigated.

Objective. The primary aim of this study was to assess aspects of hand function and manual capacity that influence bimanual performance in children with congenital hand differences (CHDs), ranging from surgically corrected polydactyly or syndactyly to radial dysplasia. A secondary aim was to assess whether the number of items on the Prosthetic Upper Extremity Functional Index (PUFI) can be reduced without losing information on bimanual performance in this population.

Design. A cross-sectional design was used.

Methods. One hundred six 10- to 14-year-old children with CHD participated in the study, which was conducted in a university hospital's outpatient clinic. Bimanual performance was evaluated with child self-reports on an adapted version of the PUFI, calculating ease of performance and actual use of the affected hand. Additionally, hand function and manual capacity were assessed.

Results. The median score on ease of performance was high, and, on average, the children used their affected hand actively in 97% of all activities. Manual capacity of the nondominant hand and lateral pinch strength of the dominant hand predicted attainment of maximum PUFI scores. Nonmaximum PUFI scores were predicted by opposition strength of the nondominant hand and lateral pinch strength of the dominant hand. In addition, in this patient group, only 6 items of the PUFI explained all variance in PUFI scores.

Limitations. The generalizability of the results is limited by the carefully selected age range. Second, the cross-sectional design of the study limits statements on causality on the relationships found.

Conclusion. Children with a CHD generally have good bimanual performance and, on average, perform activities with active use of the affected hand. Therapy directed toward increasing manual capacity and finger muscle strength might assist in improving bimanual performance in children with CHD. Furthermore, the number of items on the PUFI could be reduced from 38 to 6 items in children with CHD.

M.S. Ardon, PT, MSc, Department of Rehabilitation Medicine, Erasmus University Medical Center, PO Box 2040, 3000 CA Rotterdam, the Netherlands. Address all correspondence to Ms Ardon at: m.ardon@erasmusmc.nl.

W.G.M. Janssen, MD, PhD, Department of Rehabilitation Medicine, Erasmus University Medical Center.

S.E.R. Hovius, MD, PhD, Department of Plastic and Reconstructive Surgery and Hand Surgery, Erasmus University Medical Center.

H.J. Stam, MD, PhD, Department of Rehabilitation Medicine, Erasmus University Medical Center.

M. Murawska, MSc, Department of Biostatistics, Erasmus University Medical Center.

M.E. Roebroek, PhD, Department of Rehabilitation Medicine, Erasmus University Medical Center.

R.W. Selles, PhD, Department of Rehabilitation Medicine and Department of Plastic and Reconstructive Surgery and Hand Surgery, Erasmus University Medical Centre.

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Everyday functioning and development of children with congenital hand differences (CHDs) is monitored through childhood and adolescence.¹ Upper limb functioning is commonly assessed using the framework of the World Health Organization's *International Classification of Functioning, Disability and Health: Child and Youth Version* (ICF-CY),² which describes the levels of (1) body function and structures, (2) daily activities, and (3) participation. In children with CHD, impairments in body functions include, among other things, declined range of motion (ROM), muscle weakness, and diminished coordination. Limitations in basic activities, such as grasping, manipulating, and releasing of objects, being essential components of most daily activities involving the upper limb, may result in limitations in daily activities and participation.

To obtain a complete representation of the child's limitation in daily activities, a distinction needs to be made between capacity and performance.² *Capacity* describes what a child can do in a standardized environment, and *performance* describes what a child does do in his or her current environment. What a child does do in daily activities is influenced not only by what he or she can do but also by the physical and social contexts.² In the hand activity litera-

ture, capacity is mostly referred to as manual capacity or manual ability, whereas performance is commonly referred to as manual performance, either unimanual or bimanual.

In a previous article,³ we reported that children with CHD generally reach moderate to good manual capacity and the relationship of manual capacity with hand functions (at the *International Classification of Functioning, Disability and Health* [ICF] level of body function and structures) is more distinct in non-dominant hands than in dominant hands. In the present study, we broadened our focus toward activity performance of children with CHD. Limitations in activity performance generally become more evident in performing bimanual activities because children who are unilaterally affected will perform unimanual activities with their unaffected hand⁴ and children who are bilaterally affected will choose their most able hand to perform the activity. However, because many daily tasks require cooperative use of both hands, bimanual performance can be regarded a more discriminative measure of limitations in performance of children with unilateral and bilateral CHD.

Although minimizing limitations in daily activities is a prominent goal in treating children with CHD, most surgical techniques intervene at the level of body functions of the upper limb, also referred to as hand functions (eg, strength, ROM), aiming to improve manual capacity and activity performance. This approach assumes relationships among hand functions, manual capacity, and activity performance, but the nature of these relations and evidence for hand function variables predicting performance have scarcely been investigated.⁵

Therefore, the primary aim of the present study was to investigate the aspects of hand function and manual capacity that influence bimanual performance in children with CHD. As Wright et al⁶ and Buffart and colleagues^{5,7,8} reported that the adapted Prosthetic Upper Extremity Functional Index (PUFI) has the most optimal reliability and validity to assess bimanual performance in children with CHD, we used this questionnaire to evaluate activity performance. The PUFI, however, is an extensive questionnaire consisting of 38 items. Although all items are used to evaluate performance and prosthetic use in children with an upper limb reduction deficiency, the extent to which all items contribute to evaluation of bimanual performance in children with CHD has never been investigated. Therefore, a second aim of this study was to investigate to what extent the extensive list of 38 items is needed to adequately assess bimanual performance in these patients or whether there is a potential for decreasing the number of items.

Method

Participants

Children (N=120) aged between 10 and 14 years were recruited from a database of children with a CHD treated at our hospital. Exclusion criteria were: cognitive or developmental delay and insufficient knowledge of the Dutch language. A total of 538 children (295 boys and 243 girls) met these criteria, and we randomly selected 300 participants using a computer-generated random sequence. They were invited by mail, and 120 children were willing to participate in the study. We found no differences between participants and nonparticipants regarding sex, diagnosis, and severity of the CHD. Due to missing values for some outcome measures, we were able to evaluate 106 of the children for the research purposes.



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- **eTable:** Manual Capacity: Tasks and Scoring System
- **eFigure:** The Number of Hands Per Limitation Category on Manual Activity Capacity for Dominant and Nondominant Hands of Children Who Were Unilaterally and Bilaterally Affected

Table 1.Characteristics of the 10- to 14-Year-Old Participating Children (n=106) and Nonparticipating Children (n=194)^a

Characteristic	PUFI Score=100	PUFI Score <100	Participants	Nonparticipants
No. of children	28	78		
Age (y), $\bar{X} \pm SD$	11.8 $\pm$ 1.6	11.9 $\pm$ 1.7	11.8 $\pm$ 1.4	11.8 $\pm$ 1.7
Sex				
Boys	60%	53%	55%	56%
Girls	40%	47%	45%	44%
Severity of disease indicators				
Affected side				
Unilateral				
Dominant affected	10%	6%	8%	*
Nondominant affected	61%	62%	61%	*
Bilateral	29%	32%	31%	*
No. of affected digits				
1	54%	20%	29%	*
2	14%	10%	11%	*
3	7%	14%	12%	*
4	7%	12%	10%	*
5	18%	44%	37%	*
Surgical treatment				
None	14%	86%	38%	*
1 or more	41%	59%	62%	*
Diagnosis according to IFSSH classification				
Failure of formation	4%	33%	25%	24%
Failure of differentiation or separation of parts	39%	14%	21%	23%
Duplication	32%	17%	21%	22%
Overgrowth	4%	0%	1%	1%
Undergrowth	21%	32%	29%	27%
Congenital constriction ring syndrome	0%	4%	3%	3%
Generalized skeletal abnormalities	0%	0%	0%	0%

^a Data indicated by asterisks could not be retrieved validly from the patient files. PUFI=Prosthetic Upper Extremity Functional Index, IFSSH=International Federation of Societies for Surgery of the Hand (Swanson AB, Swanson GD, Tada K. A classification for congenital limb malformation. *J Hand Surg Am.* 1983;8(5 pt 2):693–702).

Parents of all children gave their informed consent for participation, as did all children above 12 years of age. Characteristics of the participating children are presented in Table 1.

Measurements

Bimanual performance of activities was assessed using the older child version (ages ≥ 7 years) of the PUFI.⁶ This questionnaire, which was originally developed for children with

transverse reduction limb deficiencies, was slightly adapted by Buffart and colleagues^{5,7,8} to enable assessment and scoring of children with diverse forms of CHD and indicated good validity (construct validity: $r = -.64$) and reliability (test-retest reliability: intraclass correlation coefficient = .83) in children with longitudinal radial deficiency. It evaluates to what extent a child actually uses the hand for 38 daily bimanual activities and is scored on a 5-point ordi-

nal scale ranging from “active use of the hand” to “cannot do” (actual use). Additionally, all 38 activities are scored on ease of activity performance, ranging from “no difficulty” to “cannot do” (ease of performance).⁸ Although the actual use can only be expressed as a proportion, the answers for ease of performance provide scaled sum scores ranging from 0 to 100 points, where higher scores indicate more ease of performance. In line with previous

Table 2.Performance of Functional Activities^a

Measure	Median (IQR)	Range
PUFI		
Ease of performance	95 (87–100)	32–100
Method of use of affected hand (%)		
Active	97 (80–100)	10–100
Passive	0 (0–4)	0–67
Forearm, elbow, trunk	0 (0–0)	0–34
One-handed	0 (0–0)	0–27
Some help	0 (0–0)	0–14
Cannot do	0 (0–0)	0–59

^a IQR=interquartile range, PUFI=Prosthetic Upper Extremity Functional Index.

research of Buffart et al,⁵ we addressed ease of performance as the primary outcome measure. In children who were bilaterally affected, we asked them to answer the questions for their most affected hand.

Manual capacity was tested according to Eliasson et al,⁹ based on the Sollerman test,¹⁰ evaluating 6 types of grip (eTable, available at ptjournal.apta.org). The test consists of 9 tasks that require handling objects with both grasp and pinch grip, scored on a 5-level ordinal scale. The scores range from 0 if the child cannot grip the object to 4 if the child can grip the object and completes the task with a normal grip and motion. All scores are added and provide a sum score between 0 and 36.

Hand function was assessed addressing joint mobility and muscle strength. We measured joint mobility with a finger goniometer to calculate the total active range of motion (TAROM) per hand¹¹ and with the Pollexograph (Erasmus University Medical Centre, Rotterdam, the Netherlands; <http://www.pollexograph.com>) to measure thumb palmar abduction.^{12,13} We calculated the TAROM per hand as a sum of all individual fingers (metacarpophalan-

geal, proximal interphalangeal, and distal interphalangeal joints) and the thumb (metacarpophalangeal and interphalangeal joints). We assessed grip and pinch strength (tip-tip pinch, tripod pinch, and lateral pinch) with the Lode handgrip and pinch grip dynamometer (Lode Medical Technology, Groningen, the Netherlands) and thumb opposition strength with the Rotterdam Intrinsic Hand Myometer (RIHM) (Erasmus University Medical Centre). For muscle strength, the mean of 3 maximum voluntary contractions was recorded. All dynamometer measurements were found to be reliable in children^{14,15} and have previously been used in children with CHD.¹⁶

A hand therapist (M.S.A.) with more than 10 years of experience with these types of measurements in this patient group performed all measurements (eg, PUFI, manual capacity, joint ROM, and muscle strength measurements).

Data Analysis

Results on centrality and spread of PUFI scores, manual capacity, and hand functions are displayed in Tables 2 and 3. In addition, the percentage of the reference value is given for manual capacity and grip, pinch, and opposition strength.^{17,18} Frequency tables on manual capacity

are presented separately for unilaterally and bilaterally affected children (eFigure, parts A and B, respectively, available at ptjournal.apta.org).

The distribution of all PUFI scores was skewed to the left, which is typical for the bounded responses. Twenty-eight children scored maximally on the PUFI, and the remaining part of the sample was normally distributed. To account for the skewed distribution of scores, we built separate models to identify factors associated with bimanual performance. Three models (Tab. 4) were constructed to determine predictors of maximal PUFI scores versus non-maximal PUFI scores. In this binary logistic regression analysis, original PUFI scores were transformed in 1 (PUFI score=100; n=28) and 0 (PUFI score <100; n=78), and we modeled to what extent PUFI scores were determined by hand functions (hand function model), manual capacity (manual capacity model), or manual capacity and hand functions combined into 1 model (combined model). Consecutively, the same models were constructed to analyze the subgroup of original PUFI scores less than 100 (n=78) using simple linear regression analysis with the same covariates. All models were built following the stepwise forward procedure.

In all models, we corrected for severity of the CHD using covariates of unilateral or bilateral affected and the number of affected digits per hand (1–5). Except for the PUFI questionnaire, all measurements were taken for both hands separately, referred to as dominant and nondominant hands.

Due to the maximal score on a large number of the 38 items of the PUFI, we decided to analyze the possibility of reducing the number of items to those that were most discriminating. To achieve this reduction, we built

a regression tree using recursive partitioning for the PUFI response.

We have applied an analysis of variance as a splitting method with a complexity parameter ($cp=0.5$) to prune off the irrelevant splits. We used rpart package in R (R Foundation for Statistical Computing, Vienna, Austria). All other statistical analyses were performed using the IBM SPSS software package for Windows version 20.0 (IBM Corp, Armonk, New York).

Role of the Funding Source

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Results

The median PUFI score on ease of performance was high (Tab. 2); approximately 96% of all activities could be performed independently, whereas only 1% of the activities were performed with help of someone else and 3% of the activities could not be performed. Children in our group scored high on ease of performance, and, on average, the children performed 97% of bimanual activities with active use of the affected hand.

Table 4 displays the results of the regression models predicting the maximum and nonmaximum PUFI scores. Only factors that significantly contributed to the model ($P<.05$) are shown. In the regression model with manual capacity, manual capacity of the nondominant hand significantly contributed to predicting the maximal PUFI score (odds ratio [OR]=1.14). In the regression model using hand functions (ROM and strength variables) as predictors, only grip strength (OR=1.03) was associated with attainment of a maximum PUFI score. The OR of 1.03

Table 3.
Manual Capacity and Body Functions Assessment^a

Measure	Median (IQR)	Possible Range
Manual capacity		
Dominant hand	36 (35–36)	0–36
Nondominant hand	28 (17–36)	0–36
	$\bar{X}$ (SD)	% of Reference Value ^{13,17,18} $\bar{X}$ (SD)
Hand functions		
Mobility (°)		
TAROM		
Dominant hand	911 (344)	NA
Nondominant hand	745 (508)	
Palm abduction		
Dominant hand	42 (19)	89 (39)
Nondominant hand	35 (23)	71 (47)
Strength (N)		
Grip		
Dominant hand	165 (84)	87 (37)
Nondominant hand	90 (71)	49 (36)
Tip-tip		
Dominant hand	31 (16)	73 (33)
Nondominant hand	20 (16)	47 (36)
Tripod		
Dominant hand	42 (20)	75 (32)
Nondominant hand	22 (22)	40 (39)
Key		
Dominant hand	51 (25)	79 (33)
Nondominant hand	31 (26)	48 (39)
Opposition		
Dominant hand	54 (19)	85 (28)
Nondominant hand	36 (27)	59 (43)

^a IQR=interquartile range; TAROM=total active range of motion; NA=not applicable, as no reference values are available.

means that the odds of scoring 100 on the PUFI's ease of performance scale were 1.03 times higher ($e^{0.029}$) for a child who, at a given level of all other covariates we controlled for in the logistic regression models, scored 1 unit higher on the scale of grip strength. When both hand function measures and manual capacity scores were entered into the model (model 3), manual capacity of the nondominant hand (OR=1.02) and lateral pinch strength (OR=1.15) of

the dominant hand predicted the maximum PUFI score.

The regression model predicting the spread in nonmaximum PUFI scores based on manual capacity scores did not reveal contributing factors, and the regression model with hand functions solely and combined with manual capacity gave similar results (Tab. 5). In both models, opposition strength of the nondominant hand ($\beta=.090$) together with lateral pinch

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Table 4.

Logistic Regression Models Identifying Predictors of Maximum and Nonmaximum Prosthetic Upper Extremity Functional Index (PUFI) Scores^a

Model	Variables in the Model	Nagelkerke R^2	Variables Contributing to the Model	β	SE	OR (95% CI)	P
Hand functions model	Hand functions (dominant + nondominant hands) Severity	.235	Grip strength, nondominant hand	.029	.007	1.03 (1.01–1.05)	<.001
Manual capacity model	Manual capacity (dominant + nondominant hands) Severity	.267	Manual capacity, nondominant hand	.128	.036	1.14 (1.06–1.22)	<.001
Combined model	Hand functions (dominant + nondominant hands) Manual capacity (dominant + nondominant hands) Severity	.322	Lateral pinch strength, dominant hand	.018	.009	1.02 (1.00–1.04)	.038
			Manual capacity, nondominant hand	.137	.036	1.15 (1.07–1.23)	<.001

^a Hand function variables included in the model were total active range of motion, thumb palmar abduction, grip pinch, tip-tip pinch, tripod pinch, and key pinch strength. Severity of disease was expressed as number of affected digits and unilateral or bilateral involvement. SE=standard error, OR=odds ratio, 95% CI=95% confidence interval.

strength of the dominant hand ($\beta=.086$) explained 19% of the variance in nonmaximum PUFI scores, whereas manual capacity did not contribute significantly to the last model. A β of .090 means that, among the subgroup of children who did not achieve maximum scores on the PUFI, but who were at the same level with respect to all covariates controlled for in the regression model, a 1-unit difference in the affected hand's opposition

strength is associated with a PUFI score that is 0.09 points higher.

The regression tree for PUFI items is displayed in the Figure, showing that only 6 items on the PUFI explain all variance in scoring. The items were peeling fruit, putting on a pair of jeans, using a can opener, tying shoelaces, putting on mittens, and putting on a necklace.

Discussion

In this study, we investigated bimanual performance measured with the child version of the PUFI in children with CHD. The children scored high on ease of performance; on average, they performed 97% of the bimanual activities with active use of the affected hand. We also investigated whether bimanual performance is predicted by hand functions (eg, mobility, strength), manual capacity (the ability to handle objects with

Table 5.

Linear Regression Models Identifying Predictors of Spread in Nonmaximum Prosthetic Upper Extremity Functional Index (PUFI) Scores^a

Model	Variables in the Model	R^2	Variables Contributing to the Model	β (95% CI)	SE	P
Hand functions model	Hand functions (dominant + nondominant hands) Severity	.191	Opposition strength, nondominant hand	.090 (.037–.144)	.027	.001
			Lateral pinch strength, dominant hand	.086 (.018–.155)	.034	.014
Manual capacity model	Manual capacity (dominant + nondominant hands) Severity		No variables			
Combined model	Hand functions (dominant + nondominant hands) Manual capacity (dominant + nondominant hands) Severity	.191	Opposition strength, nondominant hand	.090 (.037–.144)	.027	.001
			Lateral pinch strength, dominant hand	.086 (.018–.155)	.034	.014

^a Hand function variables included in the model were total active range of motion, thumb palmar abduction, grip pinch, tip-tip pinch, tripod pinch, and key pinch strength. Severity of disease was expressed as number of affected digits and unilateral or bilateral involvement. SE=standard error, 95% CI=95% confidence interval.

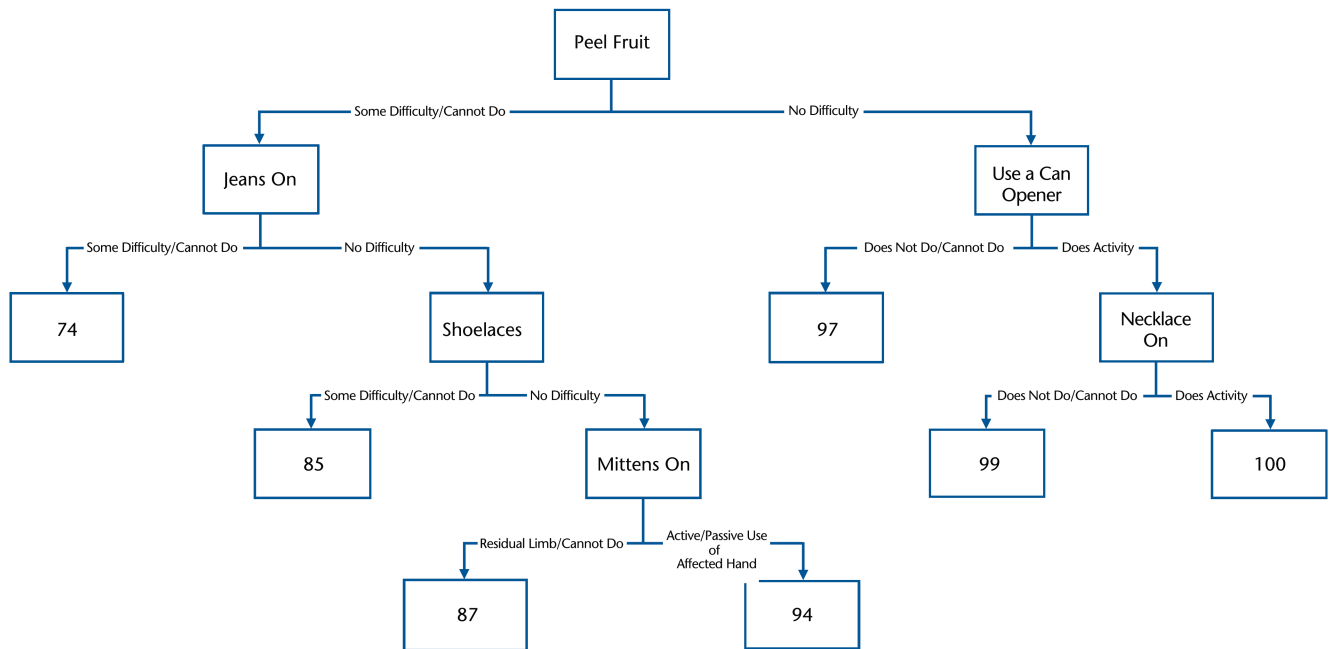


Figure.

Regression tree for Prosthetic Upper Extremity Functional Index (PUFI) items. The first discriminating item is “peel fruit.” If children answer that they experience no difficulty, the next item is “Use a can opener” and so on. This process results in an average ease of performance score; the numbers in the boxes are actual PUFI scores on ease of performance.

grasp and pinch grips), or both. Due to a skewed distribution of the outcome, we differentiated between predicting maximal PUFI score versus nonmaximal PUFI score, followed by predicting spread in the nonmaximal PUFI group. Overall, when combining both models, bimanual performance was predicted by 2 measures of muscle strength (eg, opposition strength of the nondominant hand and lateral pinch strength of the dominant hand) and by manual capacity of the nondominant hand. We also evaluated whether all 38 PUFI items are relevant in a group of children with different CHD. We found only 6 discriminating items, indicating that for future use in this patient group, the number of PUFI items could be reduced.

Despite the PUFI having the most optimal reliability and validity to assess bimanual performance in children with CHD,⁵⁻⁸ the relatively

high scores found in this study may suggest that this measure is not sensitive enough to detect problems with bimanual activities in children with different kinds of CHD. The high scores may result from a lack of sensitivity, but they also may indicate that children with CHD in this age group do not feel many restrictions in performing their daily activities.

We found that manual capacity of the nondominant hand was associated with the attainment of maximal scores on bimanual performance, which is comparable to the results of Sakzewski et al¹⁹ in children with cerebral palsy (CP) and may have implications for intervention strategies in training children with CHD. Currently, in rehabilitation, 2 contrasting intervention strategies for enhancing use of an affected hand in daily activities exist: constraint-induced movement therapy (CIMT)^{20,21} (ie, forced use of the

affected hand) and bimanual intensive therapy (BIT).²² Recently, Dong et al²³ reported that CIMT improved unimanual capacity of the impaired arm more than BIT but that children may improve more in both bimanual performance and self-determined overall life goals following BIT. Therefore, therapists already use a combination of CIMT and BIT to improve arm function for children with unilateral CP. The findings in this study suggest that it may be valuable to further study the use and effectiveness of these interventions in children with CHD.

Bimanual activity performance also was predicted by grip strength and opposition strength of the nondominant hand and lateral pinch strength of the dominant hand in children who were not achieving a perfect score of 100 on the PUFI. Although grip strength of the nondominant hand accounted for 24% of the variance in the first model, it was not a

significant contributor in the final model when manual capacity was added. Similarly, Sakzewski et al¹⁹ found no relationship between weakened strength and bimanual performance in children with hemiplegic CP. In contrast, Arnould et al²⁴ did find a relationship between grip strength of the nondominant hand and activity performance in children with CP.

The association of opposition strength of the nondominant hand with non-maximal scoring on bimanual performance underlines the importance of opposition strength in children with CHD. In this patient group, hand surgeons perform opponensplasties, which are known to increase opposition strength in thumbs with weakened or lacking opposition strength.^{25,26} Our finding is in line with a study in children with Charcot-Marie-Tooth disease. In that study, an association was demonstrated between gain in opposition strength and gain in manual capacity and performance.²⁷

We found that bimanual activity performance was predicted by variables from both the dominant hand and the nondominant hand, which underscores that bimanual daily activities require different roles for each hand. Therefore, in children with CHD, each hand may be a limiting factor. In children with CP, Arnould et al²⁴ also found that different qualities of movement are addressed in the dominant hand and the nondominant hand in bimanual activities. They stated that “the achievement of manual activities requires a highly dexterous dominant hand and a strong and an enough dexterous nondominant hand to ensure adjustable stabilization of the objects.”^{24(p712)} This finding is in accordance with our results that hand strength and manual capacity of the nondominant hand did predict bimanual performance

in our group. In contrast to children with CP, children with CHD showed sufficient levels of manual capacity of the dominant hand, which did not predict bimanual performance in children with CHD.

Despite a maximal unimanual capacity in both hands, some children in our group did not score maximally on bimanual performance. This finding suggests that not only capacity or hand function determines bimanual performance. It supports the ICF-CY theory that relationships between function and activities are not straightforward and that personal factors as well as environmental factors play a role.² Because the child's personal factors (eg, motivation, adaptability) and factors concerning school, family, and social environment were not studied, future research on the influence of these factors is still needed.

A second aim of our study was to assess whether all 38 items of the PUFI are relevant to measurement of bimanual performance in children with CHD. We found only 6 discriminating items, indicating that for future use in this patient group, the number of PUFI items could be reduced. The PUFI was originally constructed and validated for children with transversal reduction deficiencies, but it also was adapted and found to be reliable for children with CHD.^{5,7} Children in our group, with CHD of different severities, however, scored relatively high on bimanual performance. We would like to emphasize, therefore, that the regression tree that indicated that only 6 items were relevant was built with data from the children with generally high performance. In children with more activity limitations, the full PUFI may still be required to capture the complexity of their functional limitations.

In our group, we found that the number of PUFI items can be reduced to 6 because only these 6 items discriminate among performance levels in children with CHD. This reduction of items improves clinical applicability because children need less time to fill out the questionnaire. In order to improve the clinical applicability and interpretation of the results on the PUFI, a future Rasch analysis of the items using a larger group of children with CHD would be beneficial. As a first indication that a reduced-item PUFI and the full PUFI are comparable, we found that the outcomes on the old and new scores correlated well ($r = .81$, $R^2 = .66$).

In this study, bimanual performance was measured while participants performed activities with both hands simultaneously. To date, it is uncommon in rehabilitation medicine to focus on multitasking with both hands, although bimanual activities may be the primary activities that differentiate children with 2 unimpaired hands from children with either unilateral or bilateral CHD. In performing bimanual activities, children with CHD have developed alternate strategies and can choose the acting hand and stabilizing hand depending on the activity. However, when both hands are supposed to be occupied with 2 different tasks, such as holding a mobile telephone with one hand and opening the door with the other hand, both hands need to be acting hands. Hypothesizing that, in these situations, children with CHD are at a disadvantage, simultaneous task performance might need to be addressed more extensively in future studies.

Limitations of our study also need to be addressed. The age range of 10 to 14 years limits the generalizability of our results. However, at the same time, this range was carefully chosen because this is the average onset of puberty and because in the Nether-

lands there is a transition in this age group from primary school to secondary school. In addition, due to the cross-sectional design of our study, statements on causality between, for instance, opposition strength and bimanual performance cannot be made and should be confirmed in longitudinal studies or randomized clinical trials.

In conclusion, the present study showed that children with a CHD generally have good bimanual performance and that, on average, activities are performed with active use of the affected hand. Bimanual performance is associated with manual capacity and muscle strength. Non-maximal scores on bimanual performance are predicted by opposition strength of the nondominant hand and lateral pinch strength of the dominant hand, suggesting that for improvement of bimanual performance, intervening at these items would be most beneficial. Furthermore, we conclude that the number of items of the PUFU questionnaire could potentially be reduced, as we found that variation in scores can be explained by only 6 items in children with CHD.

Ms Ardon, Dr Janssen, Dr Hovius, Dr Roebroek, and Dr Selles provided concept/idea/research design. Ms Ardon, Dr Janssen, Dr Roebroek, and Dr Selles provided writing. Ms Ardon provided data collection. Ms Ardon, Ms Murawska, Dr Roebroek, and Dr Selles provided data analysis. Dr Stam and Dr Selles provided project management. Dr Stam provided fund procurement. Dr Hovius provided study participants. Dr Hovius and Dr Stam provided facilities/equipment. Dr Janssen and Dr Hovius provided institutional liaisons. Dr Janssen, Dr Hovius, Dr Stam, Dr Roebroek, and Dr Selles provided consultation (including review of manuscript before submission).

The Medical Ethics Committee of Erasmus University Medical Center approved the study.

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Monique S. Ardon, Wim G.M. Janssen, Steven E.R. Hovius, Henk J. Stam, Magdalena Murawska, Marij E. Roebroek and Ruud W. Selles

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